

COURSE CODE: CSE 4204 || COURSE TITLE: MOBILE COMPUTING LAB

WEEK 01 TASK: TEAM FORMATION & PROJECT SELECTION

TEAM NAME: CSE4204-8D-T01 || SECTION: 8D

Initial Project Title: NUBTK_COPILOT - AI-Powered University ERP and Career Architect

1. Team Information & Roles

Full Member Name	Student ID	Role & core Responsibilities
Bashir Alam	11220320963	Django core backend architecture, workflow automation engines, and Open Router API/ML model deployment orchestration.
Tisha Rani Chakroborty	11220320954	Frontend layout development, interactive user dashboards, and responsive administrative view implementations via Tailwind CSS.
Fariha Rahman Oishee	11220320967	Database schema orchestration, structured relational table optimizations, and PostgreSQL management within Sup abase environment.
Ratul Raha Shuvo	11220320949	Payment gateway sub-system integration, transactional PDF generation workflows, and automated test-suite orchestration.

2. Preliminary Project Description

NUBTK_COPILOT is a next-generation Academic Resource Planning (ERP) platform and automated career architecture framework tailored for modern computing environments. Standard legacy university management infrastructures restrict their internal capabilities purely to passive administrative operations, such as tracking static historical grade registers and aggregate classroom attendance data. They conspicuously fail to provide granular tracking metrics for identifying the foundational practical skill inventories that technical software engineering students sequentially acquire over the course of an academic term (e.g., granular competencies like 'Python Loop Optimization' or 'Relational SQL Join Logic').

This system actively bridges the severe structural misalignment existing between classical academic training regimes and live industrial software recruitment expectations. By formalizing a comprehensive technical skill logging repository directly linked to operational semester course checkpoints, NUBTK_COPILOT transforms the traditional transcript into an actionable, dynamic career portfolio. Furthermore, the infrastructure seamlessly coordinates real-time student performance profiling, allowing administrative supervisors and academic advisors to pinpoint exactly where individual student tracks fall below baseline expectations.

3. Justification for Meaningful AI Integration

Artificial Intelligence acts as the foundational core engine of NUBTK_COPILOT rather than an elective utility feature. The concrete integrations include:

- **Contextual AI Mentor Hub (Open Router / Claude 3):** Integrates an advanced LLM framework directly into the student ecosystem to provide immediate semantic code diagnostics, systemic debugging roadmaps, and domain-specific career architectural paths. It operates with custom institutional boundaries to ensure academic integrity while sustaining active engagement through gamified point distributions.
- **Predictive Admin Insight Engine (Scikit-Learn):** Departs from traditional post-mortem academic reporting by employing predictive machine learning algorithms. The engine continuously runs statistical analyses on aggregate course evaluation data, early-stage exam sub-scores, and engagement logs to flag struggling students proactively before term conclusions, generating automated alerts for department chairs.
- **Algorithmic Career Readiness Scoring:** Computes a complex, weighted indexing score (0–100) that balances an individual student's empirical cumulative GPA with live, verified technical competencies. The resulting matrix maps directly against live, programmatic scrapings of

programmatic marketplace needs (e.g., BD Jobs, LinkedIn API frameworks) to suggest target professional specializations and pinpoint explicit technical skill deficiencies.

4. High-Level Modular System Architecture

The abstract architectural organization of the full-stack system is structurally structured as follows to ensure strict decoupled development across backend APIs, machine learning pipelines, and responsive frontend client layers:

nubtk_copilot/	[System Root Environment]
backend/	[Django 5.0 Core Framework Services]
core/	[Global System Settings & Routing Policies]
api/	[REST ViewSets & Standardized Data Serializers]
apps/	[Decoupled Domain Service Architectures]
academics/	[Enrollments, Secure Gateway Payments, Records]
ai_mentor/	[LLM Chat Interface & OpenRouter Integrations]
ai_ml_engine/	[Scikit-Learn Regression & Skill-Gap Matrix]
portal_web/	[Lightweight Front-End Templates & Context Views]
static/	[Compiled Tailwind Design Tokens & Asset Blobs]

5. Preliminary Core Technology Stack

Layer Component	Selected Technology	Architectural and Practical Justification
Backend Web Layer	Django 5.0 (Python 3.11)	Enforces an enterprise-grade Model-Template-View architectural paradigm. Offers built-in object-relational mapping (ORM) security guarantees, automatic request validation, and clean native REST API integrations.
Data Persistence Layer	PostgreSQL (Supabase Cloud)	Guarantees full ACID compliance for critical payment processing and enrollment transactions, while natively managing complex relational models for intricate granular student skill inventories.

Intelligent Engine Layer	OpenRouter API (Claude 3) & Scikit-Learn	Claude 3 models yield superior semantic code explanations and contextual reasoning parameters. Scikit-learn implements high-performance, deterministic local predictive student evaluation models.
Client Presentation Layer	Semantic HTML5 & Tailwind CSS	Establishes a highly performant, lightning-fast rendering engine free from heavy client-side bundle weight overheads, ensuring responsive presentation matching mobile application standard viewports.

6. Analysis of Similar Existing Frameworks

An evaluation of current educational and industry platforms indicates a major structural gap that NUBTK_COPILOT systematically resolves:

- **Standard University ERP Architectures (e.g., Camu, Ellucian Banner):** Highly functional administrative tools for payment processing, ledger management, and broad course registrations, but completely blind to the explicit, ongoing practical technical skill developments of students.
- **LinkedIn Talent Insights & Marketplace Matrix:** Maintains excellent industrial marketplace demand data mappings and aggregates job skills profiles effectively, but completely lacks structural ties to live internal university learning pathways or immediate academic coursework tracking.
- **Coursera Integrated AI Coach Ecosystem:** Provides outstanding isolated contextual code guidance and learning tracking parameters, but remains separated from institutional operational systems such as formal institutional grading, administrative compliance, and internal student support interventions.